

Project: **ETAP**
Location: **19.0.1C**
Contract:
Engineer:
Filename: Eksisting_PB_SLEMAN
Study Case: LF

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Electrical Transient Analyzer Program

Load Flow Analysis

Loading Category (2): Normal

Generation Category (1): Design

Load Diversity Factor: Global

Constant kVA=130%, Constant Z=130%, Generic=130%, Constant I=130%

	Swing	V-Control	Load	Total
Number of Buses:	1	0	18	19

	XFMR2	XFMR3	Reactor	Line/Cable/ Busway	Impedance	Tie PD	Total
Number of Branches:	2	0	0	16	0	0	18

Method of Solution: Adaptive Newton-Raphson Method

Maximum No. of Iteration: 99

Precision of Solution: 0.0001000

System Frequency: 50.00 Hz

Unit System: Metric

Project Filename: Eksisting_PB_SLEMAN

Output Filename: E:\KULIAH\SKRIPSI\Etap\GI_KENTUNGAN_SKRIPSI\SudahPecahBeban2\SudahPecahBeban2.Ifr

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Adjustments

<u>Tolerance</u>	<u>Apply Adjustments</u>	<u>Individual /Global</u>	<u>Percent</u>
Transformer Impedance:	Yes	Individual	
Reactor Impedance:	Yes	Individual	
Overload Heater Resistance:	No		
Transmission Line Length:	No		
Cable / Busway Length:	No		
<u>Temperature Correction</u>	<u>Apply Adjustments</u>	<u>Individual /Global</u>	<u>Degree C</u>
Transmission Line Resistance:	Yes	Individual	
Cable / Busway Resistance:	Yes	Individual	

Bus Input Data

Bus			Initial Voltage		Load							
					Constant kVA		Constant Z		Constant I		Generic	
ID	kV	Sub-sys	% Mag.	Ang.	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar
Bus1	150.000	1	100.0	0.0								
Bus2	20.000	1	100.0	0.0								
Bus3	20.000	1	100.0	0.0								
Bus4	20.000	1	100.0	0.0								
Bus5	20.000	1	100.0	0.0			1.216	0.753				
Bus6	20.000	1	100.0	0.0								
Bus7	20.000	1	100.0	0.0								
Bus8	20.000	1	100.0	0.0								
Bus9	20.000	1	100.0	0.0								
Bus10	20.000	1	100.0	0.0								
Bus11	20.000	1	100.0	0.0								
Bus12	20.000	1	100.0	0.0								
Bus13	20.000	1	100.0	0.0			1.430	0.886				
Bus14	20.000	1	100.0	0.0			1.116	0.692				
Bus15	20.000	1	100.0	0.0			4.593	2.847				
Bus16	20.000	1	100.0	0.0								
Bus17	20.000	1	100.0	0.0			1.125	0.697				
Bus18	20.000	1	100.0	0.0								
Bus19	20.000	1	100.0	0.0								
Total Number of Buses: 19					0.000	0.000	9.480	5.875	0.000	0.000	0.000	0.000

Generation Bus				Voltage		Generation			Mvar Limits	
ID	kV	Type	Sub-sys	% Mag.	Angle	MW	Mvar	% PF	Max	Min
Bus1	150.000	Swing	1	100.0	0.0					
						0.000	0.000			

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Line/Cable/Busway Input Data

ohms or siemens/1000 m per Conductor (Cable) or per Phase (Line/Busway)

Line/Cable/Busway		Length							
ID	Library	Size	Adj. (m)	% Tol.	#/Phase	T (°C)	R	X	Y
Cable1	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable2	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable3	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable5	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable6	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable7	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable8	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable9	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable10	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable11	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable12	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable19	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable21	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable23	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable24	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	
Cable26	23NALN3	240	1000.0	0.0	1	75	0.134400	0.346200	

Line / Cable / Busway resistances are listed at the specified temperatures.

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2-Winding Transformer Input Data

Transformer		Rating					Z Variation			% Tap Setting		Adjusted	Phase Shift	
ID	Phase	MVA	Prim. kV	Sec. kV	% Z1	X1/R1	+ 5%	- 5%	% Tol.	Prim.	Sec.	% Z	Type	Angle
TRAFO_05	3-Phase	150.000	150.000	20.000	6.75	5.79	0	0	0	0	0	6.7500	YNyn	0.000
TRAFO_06	3-Phase	150.000	150.000	20.000	12.60	5.79	0	0	0	0	0	12.6000	YNyn	0.000

Branch Connections

CKT/Branch		Connected Bus ID		% Impedance, Pos. Seq., 100 MVA Base			
ID	Type	From Bus	To Bus	R	X	Z	Y
TRAFO_05	2W XFMR	Bus1	Bus2	0.77	4.43	4.50	
TRAFO_06	2W XFMR	Bus1	Bus9	1.43	8.28	8.40	
Cable1	Cable	Bus2	Bus3	3.36	8.66	9.28	
Cable2	Cable	Bus3	Bus4	3.36	8.66	9.28	
Cable3	Cable	Bus5	Bus4	3.36	8.66	9.28	
Cable5	Cable	Bus4	Bus6	3.36	8.66	9.28	
Cable6	Cable	Bus6	Bus7	3.36	8.66	9.28	
Cable7	Cable	Bus7	Bus8	3.36	8.66	9.28	
Cable8	Cable	Bus9	Bus10	3.36	8.66	9.28	
Cable9	Cable	Bus10	Bus11	3.36	8.66	9.28	
Cable10	Cable	Bus11	Bus12	3.36	8.66	9.28	
Cable11	Cable	Bus12	Bus13	3.36	8.66	9.28	
Cable12	Cable	Bus12	Bus14	3.36	8.66	9.28	
Cable19	Cable	Bus12	Bus15	3.36	8.66	9.28	
Cable21	Cable	Bus16	Bus17	3.36	8.66	9.28	
Cable23	Cable	Bus16	Bus18	3.36	8.66	9.28	
Cable24	Cable	Bus8	Bus18	3.36	8.66	9.28	
Cable26	Cable	Bus4	Bus19	3.36	8.66	9.28	

LOAD FLOW REPORT

Bus		Voltage		Generation		Load		Load Flow					XFMR	
ID	kV	% Mag.	Ang.	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap	
* Bus1	150.000	100.000	0.0	9.174	5.909	0.000	0.000	Bus2	2.312	1.454	10.5	84.7		
								Bus9	6.862	4.455	31.5	83.9		
Bus2	20.000	99.918	-0.1	0.000	0.000	0.000	0.000	Bus3	2.311	1.450	78.8	84.7		
								Bus1	-2.311	-1.450	78.8	84.7		
Bus3	20.000	99.715	-0.1	0.000	0.000	0.000	0.000	Bus2	-2.309	-1.444	78.8	84.8		
								Bus4	2.309	1.444	78.8	84.8		
Bus4	20.000	99.512	-0.2	0.000	0.000	0.000	0.000	Bus3	-2.306	-1.437	78.8	84.9		
								Bus5	1.202	0.746	41.0	85.0		
								Bus6	1.104	0.691	37.8	84.8		
								Bus19	0.000	0.000	0.0	0.0		
Bus5	20.000	99.406	-0.3	0.000	0.000	1.201	0.744	Bus4	-1.201	-0.744	41.0	85.0		
Bus6	20.000	99.414	-0.3	0.000	0.000	0.000	0.000	Bus4	-1.104	-0.690	37.8	84.8		
								Bus7	1.104	0.690	37.8	84.8		
Bus7	20.000	99.317	-0.3	0.000	0.000	0.000	0.000	Bus6	-1.103	-0.688	37.8	84.8		
								Bus8	1.103	0.688	37.8	84.8		
Bus8	20.000	99.220	-0.4	0.000	0.000	0.000	0.000	Bus7	-1.103	-0.687	37.8	84.9		
								Bus18	1.103	0.687	37.8	84.9		
Bus9	20.000	99.534	-0.3	0.000	0.000	0.000	0.000	Bus10	6.853	4.400	236.2	84.1		
								Bus1	-6.853	-4.400	236.2	84.1		
Bus10	20.000	98.921	-0.5	0.000	0.000	0.000	0.000	Bus9	-6.830	-4.342	236.2	84.4		
								Bus11	6.830	4.342	236.2	84.4		
Bus11	20.000	98.311	-0.8	0.000	0.000	0.000	0.000	Bus10	-6.808	-4.284	236.2	84.6		
								Bus12	6.808	4.284	236.2	84.6		
Bus12	20.000	97.702	-1.1	0.000	0.000	0.000	0.000	Bus11	-6.785	-4.226	236.2	84.9		
								Bus13	1.362	0.846	47.4	85.0		
								Bus14	1.064	0.660	37.0	85.0		
								Bus15	4.359	2.720	151.8	84.8		
Bus13	20.000	97.580	-1.1	0.000	0.000	1.362	0.844	Bus12	-1.362	-0.844	47.4	85.0		
Bus14	20.000	97.607	-1.1	0.000	0.000	1.063	0.659	Bus12	-1.063	-0.659	37.0	85.0		
Bus15	20.000	97.311	-1.2	0.000	0.000	4.350	2.696	Bus12	-4.350	-2.696	151.8	85.0		
Bus16	20.000	99.025	-0.4	0.000	0.000	0.000	0.000	Bus17	1.101	0.684	37.8	85.0		

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Bus		Voltage		Generation		Load		Load Flow				XFMR	
ID	kV	% Mag.	Ang.	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap
								Bus18	-1.101	-0.684	37.8	85.0	
Bus17	20.000	98.928	-0.5	0.000	0.000	1.101	0.682	Bus16	-1.101	-0.682	37.8	85.0	
Bus18	20.000	99.122	-0.4	0.000	0.000	0.000	0.000	Bus16	1.102	0.685	37.8	84.9	
								Bus8	-1.102	-0.685	37.8	84.9	
Bus19	20.000	99.512	-0.2	0.000	0.000	0.000	0.000	Bus4	0.000	0.000	0.0	0.0	

* Indicates a voltage regulated bus (voltage controlled or swing type machine connected to it)

Indicates a bus with a load mismatch of more than 0.1 MVA

Bus Loading Summary Report

Bus			Directly Connected Load								Total Bus Load			
			Constant kVA		Constant Z		Constant I		Generic		MVA	% PF	Amp	Percent Loading
ID	kV	Rated Amp	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar				
Bus1	150.000										10.912	84.1	42.0	
Bus2	20.000										2.728	84.7	78.8	
Bus3	20.000										2.723	84.8	78.8	
Bus4	20.000										2.717	84.9	78.8	
Bus5	20.000				1.201	0.744					1.413	85.0	41.0	
Bus6	20.000										1.302	84.8	37.8	
Bus7	20.000										1.300	84.8	37.8	
Bus8	20.000										1.299	84.9	37.8	
Bus9	20.000										8.144	84.1	236.2	
Bus10	20.000										8.094	84.4	236.2	
Bus11	20.000										8.044	84.6	236.2	
Bus12	20.000										7.994	84.9	236.2	
Bus13	20.000				1.362	0.844					1.602	85.0	47.4	
Bus14	20.000				1.063	0.659					1.251	85.0	37.0	
Bus15	20.000				4.350	2.696					5.117	85.0	151.8	
Bus16	20.000										1.296	85.0	37.8	
Bus17	20.000				1.101	0.682					1.295	85.0	37.8	
Bus18	20.000										1.298	84.9	37.8	
Bus19	20.000													

* Indicates operating load of a bus exceeds the bus critical limit (100.0% of the Continuous Ampere rating).

Indicates operating load of a bus exceeds the bus marginal limit (95.0% of the Continuous Ampere rating).

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Branch Loading Summary Report

CKT / Branch		Busway / Cable & Reactor			Transformer				
ID	Type	Ampacity (Amp)	Loading Amp	%	Capability (MVA)	Loading (input)		Loading (output)	
						MVA	%	MVA	%
Cable1	Cable	585.00	78.83	13.48					
Cable2	Cable	585.00	78.83	13.48					
Cable3	Cable	585.00	41.04	7.01					
Cable5	Cable	585.00	37.79	6.46					
Cable6	Cable	585.00	37.79	6.46					
Cable7	Cable	585.00	37.79	6.46					
Cable8	Cable	585.00	236.19	40.37					
Cable9	Cable	585.00	236.19	40.37					
Cable10	Cable	585.00	236.19	40.37					
Cable11	Cable	585.00	47.39	8.10					
Cable12	Cable	585.00	37.00	6.32					
Cable19	Cable	585.00	151.81	25.95					
Cable21	Cable	585.00	37.79	6.46					
Cable23	Cable	585.00	37.79	6.46					
Cable24	Cable	585.00	37.79	6.46					
Cable26	Cable	585.00							
TRAFO_05	Transformer				150.000	2.731	1.8	2.728	1.8
TRAFO_06	Transformer				150.000	8.182	5.5	8.144	5.4

* Indicates a branch with operating load exceeding the branch capability.

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Branch Losses Summary Report

Branch ID	From-To Bus Flow		To-From Bus Flow		Losses		% Bus Voltage		Vd % Drop in Vmag
	MW	Mvar	MW	Mvar	kW	kvar	From	To	
Cable1	2.311	1.450	-2.309	-1.444	2.5	6.5	99.9	99.7	0.20
Cable10	6.808	4.284	-6.785	-4.226	22.5	57.9	98.3	97.7	0.61
Cable11	1.362	0.846	-1.362	-0.844	0.9	2.3	97.7	97.6	0.12
Cable12	1.064	0.660	-1.063	-0.659	0.6	1.4	97.7	97.6	0.10
Cable19	4.359	2.720	-4.350	-2.696	9.3	23.9	97.7	97.3	0.39
Cable2	2.309	1.444	-2.306	-1.437	2.5	6.5	99.7	99.5	0.20
Cable21	1.101	0.684	-1.101	-0.682	0.6	1.5	99.0	98.9	0.10
Cable23	-1.101	-0.684	1.102	0.685	0.6	1.5	99.0	99.1	0.10
Cable24	1.103	0.687	-1.102	-0.685	0.6	1.5	99.2	99.1	0.10
Cable26	0.000	0.000	0.000	0.000			99.5	99.5	
Cable3	1.202	0.746	-1.201	-0.744	0.7	1.7	99.5	99.4	0.11
Cable5	1.104	0.691	-1.104	-0.690	0.6	1.5	99.5	99.4	0.10
Cable6	1.104	0.690	-1.103	-0.688	0.6	1.5	99.4	99.3	0.10
Cable7	1.103	0.688	-1.103	-0.687	0.6	1.5	99.3	99.2	0.10
Cable8	6.853	4.400	-6.830	-4.342	22.5	57.9	99.5	98.9	0.61
Cable9	6.830	4.342	-6.808	-4.284	22.5	57.9	98.9	98.3	0.61
TRAFO_05	2.312	1.454	-2.311	-1.450	0.6	3.3	100.0	99.9	0.08
TRAFO_06	6.862	4.455	-6.853	-4.400	9.6	55.4	100.0	99.5	0.47
					97.5	283.8			

* This Transmission Line includes Series Capacitor.

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Alert Summary Report

% Alert Settings

	<u>Critical</u>	<u>Marginal</u>
<u>Loading</u>		
Bus	100.0	95.0
Cable / Busway	100.0	95.0
Reactor	100.0	95.0
Line	100.0	95.0
Transformer	100.0	95.0
Panel	100.0	95.0
Protective Device	100.0	95.0
Generator	100.0	95.0
Inverter/Charger	100.0	95.0
<u>Bus Voltage</u>		
OverVoltage	105.0	102.0
UnderVoltage	95.0	98.0
<u>Generator Excitation</u>		
OverExcited (Q Max.)	100.0	95.0
UnderExcited (Q Min.)	100.0	

Marginal Report

Device ID	Type	Condition	Rating/Limit	Unit	Operating	% Operating	Phase Type
Bus12	Bus	Under Voltage	20.000	kV	19.540	97.7	3-Phase
Bus13	Bus	Under Voltage	20.000	kV	19.516	97.6	3-Phase
Bus14	Bus	Under Voltage	20.000	kV	19.521	97.6	3-Phase
Bus15	Bus	Under Voltage	20.000	kV	19.462	97.3	3-Phase

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SUMMARY OF TOTAL GENERATION , LOADING & DEMAND

	MW	Mvar	MVA	% PF
Source (Swing Buses):	9.174	5.909	10.912	84.07 Lagging
Source (Non-Swing Buses):	0.000	0.000	0.000	
Total Demand:	9.174	5.909	10.912	84.07 Lagging
Total Motor Load:	0.000	0.000	0.000	
Total Static Load:	9.077	5.625	10.678	85.00 Lagging
Total Constant I Load:	0.000	0.000	0.000	
Total Generic Load:	0.000	0.000	0.000	
Apparent Losses:	0.098	0.284		
System Mismatch:	0.000	0.000		

Number of Iterations: 3